# Chapter 4. Rings and Fields

Rings, integral domains, Fermat and Euler, fields of quotients, polynomial rings, factorization, quaternions, ordered rings

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## 4.1 Rings and fields (§18)

**Motivation.** Rings abstract two operations linked by the distributive law; fields are the rings in which division is possible.

**Definition (Ring)**

A **ring** $\langle R, +, \cdot \rangle$ is a set with two binary operations such that
$\mathcal{R}_1$: $\langle R, + \rangle$ is an abelian group;
$\mathcal{R}_2$: multiplication is associative;
$\mathcal{R}_3$: both distributive laws $a(b+c) = ab + ac$, $(a+b)c = ac + bc$ hold.
$R$ is **commutative** if multiplication is; a **unity** is a multiplicative identity $1 \neq 0$; a **unit** is an element with a multiplicative inverse. A **division ring** (skew field) is a ring with unity in which every nonzero element is a unit; a **field** is a commutative division ring.

**Theorem 4.1 (18.8)**

In a ring, (1) $0a = a0 = 0$;
(2) $a(-b) = (-a)b = -(ab)$;
(3) $(-a)(-b) = ab$.

**Proof.** (1) $a0 + a0 = a(0+0) = a0 = a0 + 0$; cancel in the additive group. Similarly $0a = 0$.

(2) $ab + a(-b) = a(b + (-b)) = a0 = 0$, so $a(-b) = -(ab)$; likewise $(-a)b = -(ab)$.

(3) $(-a)(-b) = -(a(-b)) = -(-(ab)) = ab$. ■

A **ring homomorphism** $\phi : R \to R'$ satisfies $\phi(a+b) = \phi(a) + \phi(b)$ and $\phi(ab) = \phi(a)\phi(b)$; an **isomorphism** is a bijective homomorphism. Examples: $\mathbb{Z}_n$; $\mathbb{Z} \times \mathbb{Z}$; $M_n(R)$; the ring of functions $\mathbb{R} \to \mathbb{R}$; the reduction map $\mathbb{Z} \to \mathbb{Z}_n$. By the Chinese remainder argument, $\mathbb{Z}_{mn} \cong \mathbb{Z}_m \times \mathbb{Z}_n$ as rings when $\gcd(m,n) = 1$ ($k \mapsto (k \bmod m, k \bmod n)$ is a ring homomorphism, and an additive isomorphism by Theorem 2.14).

## 4.2 Integral domains (§19)

**Motivation.** In $\mathbb{Z}_6$, $2 \cdot 3 = 0$; integral domains are exactly the rings where cancellation survives.

Nonzero $a, b$ with $ab = 0$ are **divisors of** 0.

**Theorem 4.2 (19.3)**

The divisors of 0 in $\mathbb{Z}_n$ are exactly the nonzero elements not relatively prime to $n$.

**Proof.** If $m \neq 0$ and $d = \gcd(m,n) \neq 1$, then $m \cdot (n/d) = (m/d)n = 0$ in $\mathbb{Z}_n$ with $n/d \neq 0$. If $\gcd(m,n) = 1$ and $ms = 0$ in $\mathbb{Z}_n$, then $n \mid ms$, so $n \mid s$ and $s = 0$. ■

**Corollary 4.3 (19.4)**

If $p$ is prime, $\mathbb{Z}_p$ has no divisors of 0.

**Theorem 4.4 (19.5)**

The cancellation laws ($ab = ac$, $a \neq 0 \Rightarrow b = c$, and similarly on the right) hold in $R$ iff $R$ has no divisors of 0.

**Proof.** If cancellation holds and $ab = 0 = a0$ with $a \neq 0$, then $b = 0$. Conversely, if there are no divisors of 0 and $ab = ac$, $a \neq 0$, then $a(b - c) = 0$, so $b - c = 0$. ■

An **integral domain** is a commutative ring with unity and no divisors of 0.

**Theorem 4.5 (19.9)**

Every field is an integral domain.

**Proof.** If $ab = 0$ and $a \neq 0$, then $b = a^{-1}ab = 0$. ■

**Theorem 4.6 (19.11)**

Every finite integral domain is a field.

**Proof.** For $a \neq 0$, $x \mapsto ax$ is one-to-one (cancellation), hence onto the finite set $D$; so $ax = 1$ for some $x$. ■

**Corollary 4.7 (19.12)**

If $p$ is prime, $\mathbb{Z}_p$ is a field.

The **characteristic** of $R$ is the least $n \in \mathbb{Z}^+$ with $n \cdot a = 0$ for all $a \in R$, or 0 if there is none.

**Theorem 4.8 (19.15)**

Let $R$ have unity. If $n \cdot 1 \neq 0$ for all $n \in \mathbb{Z}^+$, $R$ has characteristic 0; otherwise the least $n$ with $n \cdot 1 = 0$ is the characteristic.

**Proof.** If $n \cdot 1 = 0$, then $n \cdot a = (n \cdot 1)a = 0$ for all $a$; conversely $n \cdot a = 0$ for all $a$ includes $a = 1$. ■

The characteristic of an integral domain is 0 or prime: if $n = rs$ with $1 < r, s < n$, then $(r \cdot 1)(s \cdot 1) = 0$.

## 4.3 Fermat's and Euler's theorems (§20)

**Motivation.** Lagrange's theorem on the units of $\mathbb{Z}_n$ gives Fermat's and Euler's theorems and the solvability of $ax \equiv b \pmod{m}$.

**Theorem 4.9 (20.1, little theorem of Fermat)**

If $p$ is prime and $p \nmid a$, then $a^{p-1} \equiv 1 \pmod{p}$.

**Proof.** $\mathbb{Z}_p^*$ is a group of order $p - 1$ (Corollary 4.7), so the order of $\bar{a}$ divides $p - 1$ (Theorem 2.11). ■

**Corollary 4.10 (20.2)**

$a^p \equiv a \pmod{p}$ for every integer $a$ and prime $p$.

**Proof.** Multiply Fermat by $a$; if $p \mid a$ both sides are 0. ■

**Theorem 4.11 (20.6)**

The nonzero elements of $\mathbb{Z}_n$ that are not divisors of 0 form a group $G_n$ under multiplication modulo $n$.

**Proof.** They are the $m$ with $\gcd(m, n) = 1$ (Theorem 4.2); products of such are such; $1 \in G_n$; and for $a \in G_n$, $x \mapsto ax$ is one-to-one on the finite set $G_n$ (cancellation), hence onto, so $ax = 1$ for some $x \in G_n$. ■

$\phi(n) = |G_n|$ is **Euler's phi-function**.

**Theorem 4.12 (20.8, Euler)**

If $\gcd(a, n) = 1$, then $a^{\phi(n)} \equiv 1 \pmod{n}$.

**Proof.** Lagrange in $G_n$. ■

**Theorem 4.13 (20.10)**

If $a \in \mathbb{Z}_m$ is relatively prime to $m$, then $ax = b$ has a unique solution in $\mathbb{Z}_m$ for each $b$.

**Proof.** $a$ is a unit of $\mathbb{Z}_m$; $x = a^{-1}b$, unique by cancellation. ■

**Corollary 4.14 (20.11)**

If $\gcd(a, m) = 1$, the solutions of $ax \equiv b \pmod{m}$ form exactly one residue class modulo $m$.

**Theorem 4.15 (20.12)**

Let $a, b \in \mathbb{Z}_m$ and $d = \gcd(a, m)$. Then $ax = b$ has a solution in $\mathbb{Z}_m$ iff $d \mid b$, and then it has exactly $d$ solutions.

**Proof.** If $ax = b$ in $\mathbb{Z}_m$, then $ax - b = qm$ in $\mathbb{Z}$, so $d \mid b$. Conversely, if $d \mid b$, write $a = da_1$, $b = db_1$, $m = dm_1$. In $\mathbb{Z}$, $ax - b = qm$ iff $a_1x - b_1 = qm_1$, so the solutions are the integers $x \equiv x_0 \pmod{m_1}$, where $x_0$ is the unique solution of $a_1x \equiv b_1 \pmod{m_1}$ (Theorem 4.13, as $\gcd(a_1, m_1) = 1$). Modulo $m$ these are $x_0, x_0 + m_1, \ldots, x_0 + (d-1)m_1$, which are $d$ distinct classes. ■

**Corollary 4.16 (20.13)**

$ax \equiv b \pmod{m}$ is solvable iff $d = \gcd(a, m)$ divides $b$, and then the solutions form exactly $d$ residue classes modulo $m$.

**Example 4.1**

$12x \equiv 27 \pmod{18}$: $d = 6 \nmid 27$, no solutions. $15x \equiv 27 \pmod{18}$: $d = 3$; $5x \equiv 9 \pmod{6}$ gives $x \equiv 3 \pmod{6}$, so $x \equiv 3, 9, 15 \pmod{18}$.

## 4.4 The field of quotients of an integral domain (§21)

**Motivation.** Formal fractions turn any integral domain into the smallest field containing it, as $\mathbb{Z} \subset \mathbb{Q}$.

Let $D$ be an integral domain and $S = \{(a,b) \in D \times D : b \neq 0\}$; set $(a,b) \sim (c,d)$ iff $ad = bc$.

**Lemma 4.17 (21.2)**

$\sim$ is an equivalence relation on $S$.

**Proof.** Reflexive and symmetric by commutativity. If $ad = bc$ and $cf = de$, then $adf = bcf = bde$, so $d(af - be) = 0$; as $d \neq 0$, $af = be$. ■

Let $F = S/\sim$ with classes $[(a,b)]$.

**Lemma 4.18 (21.3)**

$[(a,b)] + [(c,d)] = [(ad+bc, bd)]$ and $[(a,b)][(c,d)] = [(ac,bd)]$ are well-defined operations on $F$.

**Proof.** $bd \neq 0$. If $(a,b) \sim (a',b')$ and $(c,d) \sim (c',d')$, i.e. $ab' = a'b$, $cd' = c'd$, then $(ad+bc)b'd' = ab'dd' + bb'cd' = a'bdd' + bb'c'd = (a'd' + b'c')bd$, and $acb'd' = a'c'bd$. ■

One checks that $F$ is a field: $[(0,1)]$ and $[(1,1)]$ are the identities, $[(-a,b)]$ the additive inverse, and $[(b,a)]$ the inverse of $[(a,b)]$ for $a \neq 0$; the axioms reduce to identities in $D$.

**Lemma 4.19 (21.4)**

$i : D \to F$, $i(a) = [(a,1)]$, is an isomorphism of $D$ onto a subring of $F$.

**Proof.** $i(a) + i(b) = [(a+b,1)]$, $i(a)i(b) = [(ab,1)]$, and $[(a,1)] = [(b,1)]$ iff $a = b$. ■

**Theorem 4.20 (21.5)**

Every integral domain $D$ embeds in a field $F$ in which every element is a quotient of two elements of $D$ (a **field of quotients**).

**Proof.** Lemmas 4.17–4.19, with $[(a,b)] = i(a)i(b)^{-1}$. ■

**Theorem 4.21 (21.6)**

If $F$ is a field of quotients of $D$ and $L$ is a field containing $D$, then $\psi(a/b) = ab^{-1}$ (computed in $L$) gives an isomorphism of $F$ onto a subfield of $L$ with $\psi(a) = a$ for $a \in D$.

**Proof.** Well defined: $a/b = c/d$ means $ad = bc$, so $ab^{-1} = cd^{-1}$ in $L$. $\psi$ preserves sums and products ($ab^{-1} + cd^{-1} = (ad+bc)(bd)^{-1}$, $ab^{-1}cd^{-1} = ac(bd)^{-1}$), and it is one-to-one because a field homomorphism with $\psi(1) = 1$ has trivial kernel ($\psi(x) = 0$, $x \neq 0$, would give $1 = \psi(x)\psi(x^{-1}) = 0$). ■

**Corollary 4.22 (21.8)**

Every field containing an integral domain $D$ contains a field of quotients of $D$.

**Corollary 4.23 (21.9)**

Any two fields of quotients of $D$ are isomorphic (by an isomorphism fixing $D$).

**Proof.** If $F, F'$ are fields of quotients, $\psi : F \to F'$ of Theorem 4.21 has image containing all $ab^{-1}$, i.e. all of $F'$. ■

## 4.5 Rings of polynomials (§22)

**Motivation.** The evaluation homomorphism turns finding a root of $f$ into a question about a kernel.

A **polynomial** with coefficients in $R$ is $f(x) = \sum_{i\geq 0} a_i x^i$ with finitely many $a_i \neq 0$; polynomials are added and multiplied as usual, $(\sum a_i x^i)(\sum b_j x^j) = \sum_n (\sum_{i+j=n} a_i b_j) x^n$.

**Theorem 4.24 (22.2)**

$R[x]$ is a ring; it is commutative if $R$ is, and has unity 1 if $R$ does.

**Proof.** $\langle R[x], + \rangle$ is an abelian group (componentwise). Associativity: the coefficient of $x^n$ in $(fg)h$ and in $f(gh)$ is $\sum_{i+j+k=n} a_i b_j c_k$. Distributivity follows from that of $R$. The rest is immediate. ■

$R[x_1, \ldots, x_n] = R[x_1, \ldots, x_{n-1}][x_n]$. If $D$ is an integral domain, so is $D[x]$: the leading coefficient of $fg$ is the product of the leading coefficients, and $\deg(fg) = \deg f + \deg g$.

**Theorem 4.25 (22.4, evaluation homomorphism)**

Let $F \leq E$ be fields and $\alpha \in E$. The map $\phi_\alpha : F[x] \to E$, $\phi_\alpha(\sum a_i x^i) = \sum a_i \alpha^i$, is a ring homomorphism with $\phi_\alpha(x) = \alpha$ and $\phi_\alpha(a) = a$ for $a \in F$.

**Proof.** Additivity is clear. For products, $\phi_\alpha(fg) = \sum_n (\sum_{i+j=n} a_i b_j)\alpha^n = (\sum a_i \alpha^i)(\sum b_j \alpha^j)$ since $E$ is commutative. ■

$\alpha$ is a **zero** of $f$ if $\phi_\alpha(f) = f(\alpha) = 0$.

**Theorem 4.26 (22.11)**

$x^2 - 2$ has no zero in $\mathbb{Q}$; thus $\sqrt{2}$ is irrational.

**Proof.** If $(m/n)^2 = 2$ in lowest terms, then $m^2 = 2n^2$, so $m$ is even, $m = 2k$, $2k^2 = n^2$, and $n$ is even, a contradiction. ■

## 4.6 Factorization of polynomials over a field (§23)

**Motivation.** Over a field polynomials divide with remainder like integers, and irreducible polynomials play the role of primes.

**Theorem 4.27 (23.1, division algorithm for $F[x]$)**

Let $f, g \in F[x]$ with $g \neq 0$. There are unique $q, r \in F[x]$ with $f = gq + r$ and $r = 0$ or $\deg r < \deg g$.

**Proof.** Existence: the set $\{f - gs : s \in F[x]\}$ has an element $r$ of least degree (or 0). If $\deg r = t \geq m = \deg g$, with leading coefficients $c_t$ and $b_m$, then $r - (c_t/b_m)x^{t-m}g$ has smaller degree, a contradiction.

Uniqueness: if $gq + r = gq' + r'$, then $g(q - q') = r' - r$ has degree $< \deg g$ unless $q = q'$. ■

**Corollary 4.28 (23.3, factor theorem)**

$a \in F$ is a zero of $f$ iff $x - a$ divides $f$.

**Proof.** $f = (x - a)q + r$ with $r \in F$; evaluating at $a$, $f(a) = r$. ■

**Corollary 4.29 (23.5)**

A nonzero $f \in F[x]$ of degree $n$ has at most $n$ zeros in $F$.

**Proof.** Induction: if $a$ is a zero, $f = (x - a)q$ with $\deg q = n - 1$, and any other zero $b$ satisfies $(b - a)q(b) = 0$, so $q(b) = 0$. ■

**Corollary 4.30 (23.6)**

A finite subgroup $G$ of the multiplicative group $F^*$ of a field is cyclic. In particular the multiplicative group of a finite field is cyclic.

**Proof.** By Theorem 2.17, $G \cong \mathbb{Z}_{d_1} \times \cdots \times \mathbb{Z}_{d_r}$ with prime powers $d_i$. Let $m = \operatorname{lcm}(d_i)$; then $a^m = 1$ for all $a \in G$, so $x^m - 1$ has $|G|$ zeros and $|G| \leq m$ (Corollary 4.29). But $m \leq \prod d_i = |G|$, with equality iff the $d_i$ are pairwise relatively prime, in which case $G$ is cyclic (Corollary 2.15). ■

A nonconstant $f \in F[x]$ is **irreducible over** $F$ if it is not a product of two polynomials of lower degree in $F[x]$.

**Theorem 4.31 (23.10)**

A polynomial of degree 2 or 3 is reducible over $F$ iff it has a zero in $F$.

**Proof.** A factorization into lower degrees must contain a linear factor $ax + b$, giving the zero $-b/a$; conversely use the factor theorem. ■

**Theorem 4.32 (23.11, Gauss's lemma)**

If $f \in \mathbb{Z}[x]$ factors in $\mathbb{Q}[x]$ into polynomials of degrees $r, s < \deg f$, it factors in $\mathbb{Z}[x]$ into polynomials of the same degrees.

**Proof.** Call $h \in \mathbb{Z}[x]$ primitive if its coefficients have gcd 1. The product of primitive polynomials is primitive: if a prime $p$ divided all coefficients of $gh$, then reducing mod $p$ gives $\bar{g}\bar{h} = 0$ in the integral domain $\mathbb{Z}_p[x]$, so $\bar{g} = 0$ or $\bar{h} = 0$. Now if $f = gh$ in $\mathbb{Q}[x]$, write $g = \frac{a}{b}g_1$, $h = \frac{c}{d}h_1$ with $g_1, h_1$ primitive in $\mathbb{Z}[x]$; then $bdf = ac\, g_1 h_1$. The gcd of the coefficients of the left side is a multiple of $bd$, and that of the right side is $ac$ (as $g_1 h_1$ is primitive), so $bd \mid ac$ and $f = \frac{ac}{bd}g_1 \cdot h_1$ is a factorization in $\mathbb{Z}[x]$. ■

**Corollary 4.33 (23.12)**

If $f = x^n + a_{n-1}x^{n-1} + \cdots + a_0 \in \mathbb{Z}[x]$ with $a_0 \neq 0$ has a zero in $\mathbb{Q}$, it has a zero $m \in \mathbb{Z}$, and $m \mid a_0$.

**Proof.** A rational zero gives a linear factor in $\mathbb{Q}[x]$, hence by Theorem 4.32 a factorization $(x - m)(x^{n-1} + \cdots - a_0/m)$ in $\mathbb{Z}[x]$ (the leading coefficients multiply to 1), so $m \in \mathbb{Z}$ and $m \mid a_0$. ■

**Theorem 4.34 (23.15, Eisenstein criterion)**

Let $p$ be prime and $f = a_n x^n + \cdots + a_0 \in \mathbb{Z}[x]$ with $a_n \not\equiv 0$, $a_i \equiv 0 \pmod{p}$ for $i < n$, and $a_0 \not\equiv 0 \pmod{p^2}$. Then $f$ is irreducible over $\mathbb{Q}$.

**Proof.** Otherwise $f = (b_r x^r + \cdots + b_0)(c_s x^s + \cdots + c_0)$ in $\mathbb{Z}[x]$ with $r, s < n$. Since $p^2 \nmid a_0 = b_0 c_0$ but $p \mid a_0$, exactly one of $b_0, c_0$, say $b_0$, is divisible by $p$. As $p \nmid a_n = b_r c_s$, $p \nmid b_r$; let $i$ be least with $p \nmid b_i$, $0 < i \leq r < n$. Then $a_i = b_i c_0 + b_{i-1} c_1 + \ldots$, where $p$ divides every term except $b_i c_0$; so $p \nmid a_i$, a contradiction. ■

**Corollary 4.35 (23.17)**

The cyclotomic polynomial $\Phi_p(x) = \frac{x^p - 1}{x - 1} = x^{p-1} + \cdots + x + 1$ is irreducible over $\mathbb{Q}$ for every prime $p$.

**Proof.** $\Phi_p(x+1) = \frac{(x+1)^p - 1}{x} = x^{p-1} + \binom{p}{1}x^{p-2} + \cdots + \binom{p}{p-1}$, and $p \mid \binom{p}{k}$ for $0 < k < p$ while $\binom{p}{p-1} = p$; Eisenstein applies. A factorization of $\Phi_p(x)$ would give one of $\Phi_p(x+1)$. ■

**Theorem 4.36 (23.18)**

If $p(x)$ is irreducible in $F[x]$ and $p \mid rs$, then $p \mid r$ or $p \mid s$.

**Proof.** The set $\{ap + br\}$ is an ideal of $F[x]$, hence equals $\langle d \rangle$ (every ideal of $F[x]$ is principal, by the division algorithm as in Theorem 5.18). $d$ divides $p$, so $d$ is a unit or an associate of $p$. If $p \nmid r$, then $d$ is a unit, say $d = 1 = ap + br$; then $s = aps + brs$ is divisible by $p$. ■

**Corollary 4.37 (23.19)**

If $p$ is irreducible and divides $r_1 \cdots r_n$, then $p$ divides some $r_i$.

**Theorem 4.38 (23.20)**

Every nonconstant $f \in F[x]$ is a product of irreducible polynomials, unique up to order and nonzero constant factors.

**Proof.** Existence by induction on degree: $f$ is irreducible or a product of lower-degree factors.

Uniqueness: if $p_1 \cdots p_r = q_1 \cdots q_s$, then $p_1$ divides some $q_j$ (Corollary 4.37), and as $q_j$ is irreducible, $q_j = up_1$ with $u \in F^*$; cancel and induct. ■

## 4.7 Noncommutative examples (§24)

**Motivation.** Endomorphism rings, group rings and the quaternions are noncommutative; Wedderburn's theorem rules this out for finite division rings.

An **endomorphism** of an abelian group $A$ is a homomorphism $A \to A$; $(\phi+\psi)(a) = \phi(a)+\psi(a)$.

**Theorem 4.39 (24.1)**

The endomorphisms $\mathrm{End}(A)$ of an abelian group form a ring under addition and composition.

**Proof.** $\phi+\psi$ is an endomorphism because $A$ is abelian; $\langle \mathrm{End}(A), + \rangle$ is an abelian group with zero map and $-\phi$. Composition is associative, and $\phi(\psi+\chi) = \phi\psi + \phi\chi$ (as $\phi$ is additive) and $(\psi+\chi)\phi = \psi\phi + \chi\phi$ (by definition). ■

For a group $G$ and commutative ring $R$ with unity, the **group ring** $RG$ consists of finite formal sums $\sum a_g g$ with $(\sum a_g g)(\sum b_h h) = \sum_k (\sum_{gh=k} a_g b_h)k$.

**Theorem 4.40 (24.4)**

$RG$ is a ring.

**Proof.** Addition is componentwise. Associativity of multiplication: the coefficient of $k$ in both triple products is $\sum_{ghl=k} a_g b_h c_l$, by associativity in $G$. Distributivity follows from that in $R$. ■

The **quaternions** $\mathbb{H}$ are $a_1 + a_2 i + a_3 j + a_4 k$ ($a_i \in \mathbb{R}$) with $i^2 = j^2 = k^2 = -1$, $ij = k = -ji$, $jk = i = -kj$, $ki = j = -ik$.

**Theorem 4.41 (24.9)**

$\mathbb{H}$ is a strictly skew field.

**Proof.** The map $a+bi+cj+dk \mapsto \begin{pmatrix} a+bi & c+di \\ -c+di & a-bi \end{pmatrix}$ is an additive injection into $M_2(\mathbb{C})$ preserving the multiplication rules of $1, i, j, k$; so $\mathbb{H}$ is isomorphic to a subring of $M_2(\mathbb{C})$, and associativity and distributivity hold. For $q = a_1+a_2 i+a_3 j+a_4 k \neq 0$ and $\bar{q} = a_1 - a_2 i - a_3 j - a_4 k$, $q\bar{q} = \bar{q}q = \sum a_i^2 > 0$, so $q^{-1} = \bar{q}/\sum a_i^2$. $ij \neq ji$. ■

**Theorem 4.42 (24.10, Wedderburn)**

Every finite division ring is a field.

**Proof [Supplement].** (Witt.) Let $D$ be a finite division ring and $Z = \{z : zx = xz \ \forall x\}$ its center, a finite field of $q$ elements. $D$ is a vector space over $Z$, so $|D| = q^n$. For $x \in D$ the centralizer $C(x) = \{y : yx = xy\}$ is a division ring containing $Z$, so $|C(x)| = q^d$; as $D$ is a (left) vector space over $C(x)$, $q^n$ is a power of $q^d$, so $d \mid n$, and $d < n$ if $x \notin Z$. The class equation of $D^*$ (conjugation action, Theorem 3.20) gives

$$q^n - 1 = (q-1) + \sum \frac{q^n - 1}{q^{d_x} - 1},$$

summed over representatives of the noncentral classes, $d_x \mid n$, $d_x < n$. Let $\Phi_n(x) = \prod(x - \zeta)$ over the primitive $n$-th roots of unity $\zeta$; from $x^n - 1 = \prod_{e \mid n} \Phi_e(x)$ and induction, $\Phi_n \in \mathbb{Z}[x]$ is monic, and $\Phi_n(x)$ divides $x^n - 1$ and $(x^n-1)/(x^d-1)$ in $\mathbb{Z}[x]$ for $d \mid n$, $d < n$. Hence the integer $\Phi_n(q)$ divides $q - 1$. But for $n > 1$ each $|q - \zeta| > q - 1 \geq 1$, so $|\Phi_n(q)| > q - 1$. Therefore $n = 1$ and $D = Z$ is commutative. ■

## 4.8 Ordered rings and fields (§25)

**Motivation.** Ordered rings formalize the sign rules of $\mathbb{R}$; squares are positive, which excludes $i^2 = -1$.

A ring $R$ is **ordered** if it has a subset $P$ (positive elements) closed under addition and multiplication such that for each $a$ exactly one of $a \in P$, $a = 0$, $-a \in P$ holds.

**Theorem 4.43 (25.3)**

In an ordered ring, squares of nonzero elements are positive, the characteristic is 0, and there are no divisors of 0.

**Proof.** If $a \in P$ then $a^2 \in P$; if $-a \in P$ then $a^2 = (-a)^2 \in P$. If $ab = 0$ with $a, b \neq 0$, then $\pm a, \pm b \in P$ give $\pm ab \in P$, contradicting $ab = 0$. If $R$ has unity, $1 = 1^2 \in P$, so $n \cdot 1 \in P$ is never 0; in general $n \cdot a^2 \in P$ for $a \neq 0$. ■

**Corollary 4.44 (25.4)**

$\mathbb{Z}$ embeds in every ordered ring with unity, with its natural ordering induced, and the natural ordering is the only ordering of $\mathbb{Z}$. $\mathbb{C}$ cannot be ordered.

**Proof.** $n \mapsto n \cdot 1$ is one-to-one since the characteristic is 0, and $n \cdot 1 \in P$ for $n > 0$. In $\mathbb{Z}$, $1 = 1^2$ is positive, so all positive integers are. In $\mathbb{C}$, $-1 = i^2$ would be positive as well as 1. ■

**Theorem 4.45 (25.5)**

Defining $a < b$ iff $b - a \in P$: (trichotomy) exactly one of $a < b$, $a = b$, $b < a$ holds; (transitivity) $a < b$, $b < c \Rightarrow a < c$; (additivity) $a < b \Rightarrow a + c < b + c$; (multiplicativity) $a < b$, $0 < c \Rightarrow ac < bc$ and $ca < cb$.

**Proof.** Apply the axioms to $b-a$; $c-a = (c-b)+(b-a)$; $(b+c)-(a+c) = b-a$; $bc-ac = (b-a)c \in P$. ■

**Theorem 4.46 (25.10)**

If $R$ is ordered by $P$ and $\phi : R \to R'$ is a ring isomorphism, then $P' = \phi[P]$ orders $R'$, and $\phi(a) < \phi(b)$ iff $a < b$.

**Proof.** $\phi$ preserves sums, products and negatives and is bijective, so $P'$ satisfies the axioms; $\phi(b) - \phi(a) = \phi(b - a) \in P'$ iff $b - a \in P$. ■

**Theorem 4.47 (25.13)**

Let $D$ be an ordered integral domain with positive set $P$ and $F$ its field of quotients. Then $P' = \{a/b : ab \in P\}$ is well defined, orders $F$, and induces the order of $D$; it is the only such order on $F$.

**Proof.** If $a/b = c/d$ $(ad = bc)$, then $ab\,d^2 = (ad)(bd) = (bc)(bd) = cd\,b^2$; since $b^2, d^2 \in P$, $ab \in P$ iff $cd \in P$ (multiplicativity and trichotomy). Closure: $\frac{a}{b} + \frac{c}{d} = \frac{ad+bc}{bd}$ with $(ad+bc)bd = ab\,d^2 + cd\,b^2 \in P$, and $\frac{a}{b}\frac{c}{d} = \frac{ac}{bd}$ with $acbd = (ab)(cd) \in P$. Trichotomy for $a/b$ follows from that of $ab$. For $a \in D$, $a/1 \in P'$ iff $a \in P$.

Uniqueness: in any order of $F$ extending that of $D$, $a/b = ab \cdot (1/b)^2$ is positive iff $ab$ is. ■

So $\mathbb{Q}$ has only its natural ordering.